Proyecto del Plano Inclinado

Determinación teórica y experimental del tiempo de descenso de un bloque con fricción cinética. Reporte generado el 9/10/2026.

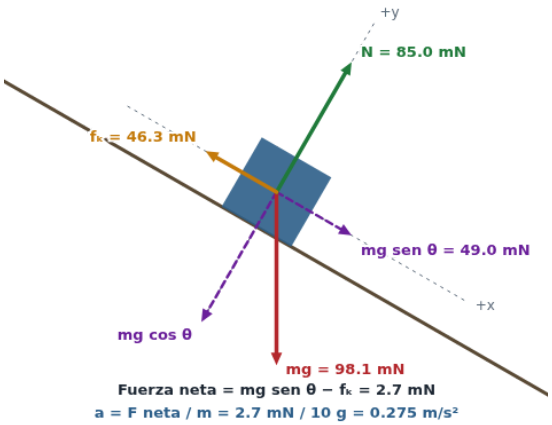
1. Parámetros

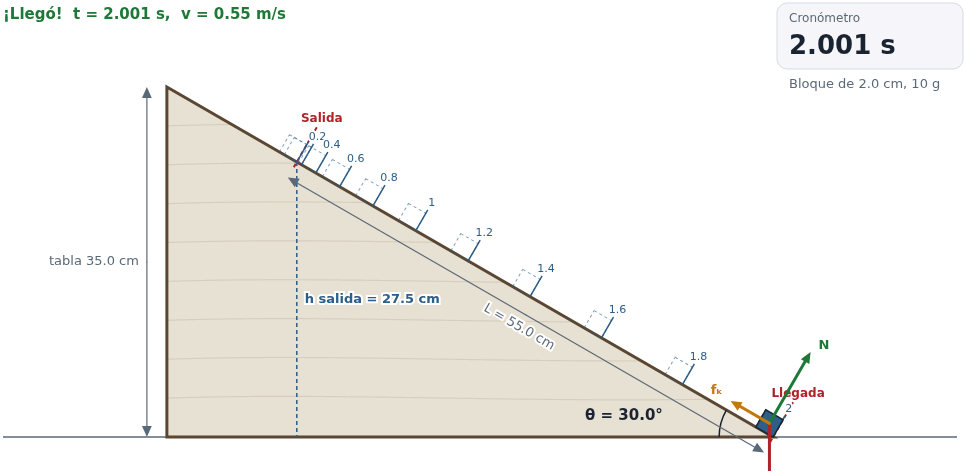
Objeto	Tamaño	m	θ	h salida	L recorrida	μ_k	g	v_0
bloque	2.0 cm	10 g	30°	27.5 cm (tabla 35.0 cm)	55.0 cm	0.545	9.81 m/s²	0 m/s

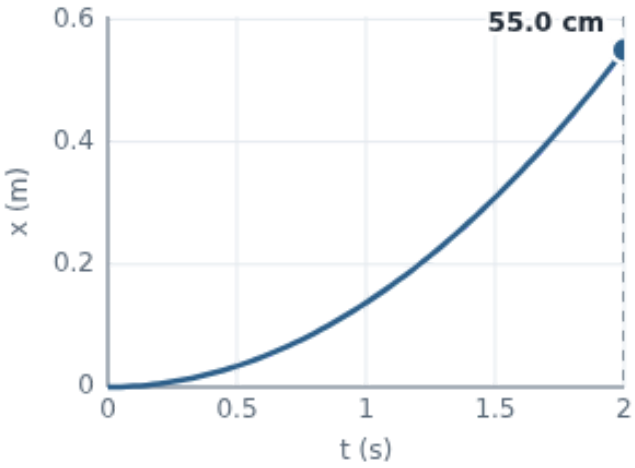
2. Análisis de fuerzas y cálculo teórico

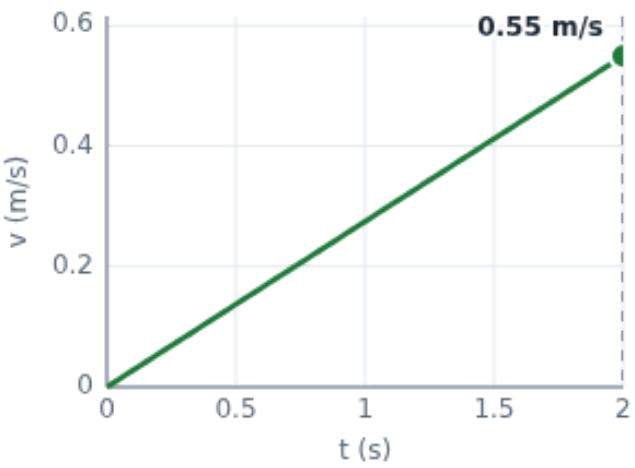
N = mg cos θ; ma = mg sen θ − μ_k mg cos θ ⟹ a = g(sen θ − μ_k cos θ) = 9.81(0.5000 − 0.4720) = **0.275 m/s²**

Recorrido L = 55.0 cm (altura de salida h = L·sen θ = 27.5 cm); t = √(2L/a) = √(2 × 0.550 / 0.275) = **2.001 s**; v = a·t = **0.55 m/s**

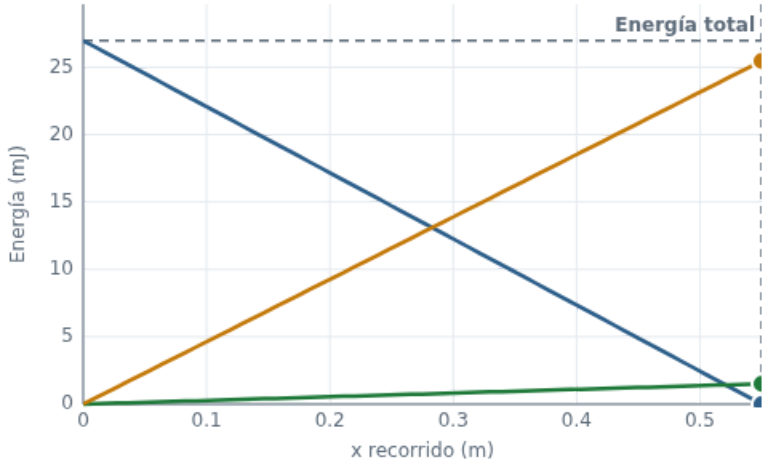





55.0 cm
0.6
0.4
0.2
0
x (m)
0
0.5
1
1.5
2
t (s)

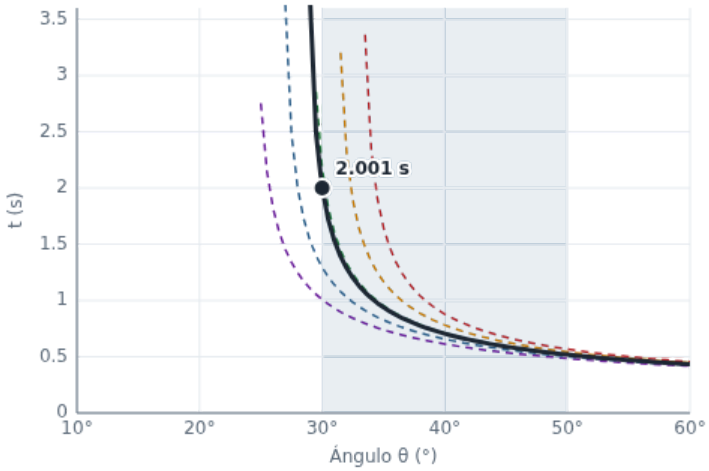

0.55 m/s
0.6
0.4
0.2
0
v (m/s)
0
0.5
1
1.5
2
t (s)

3. Energía



4. Sensibilidad (L = 55.0 cm)

θ	μ_k = 0.45	μ_k = 0.50	μ_k = 0.55	μ_k = 0.60	μ_k = 0.65
30°	1.008 s	1.294 s	**2.176 s**	no baja	no baja
35°	0.740 s	0.827 s	0.955 s	1.169 s	1.651 s
40°	0.613 s	0.657 s	0.712 s	0.782 s	0.880 s
45°	0.537 s	0.563 s	0.594 s	0.630 s	0.673 s
50°	0.485 s	0.502 s	0.521 s	0.543 s	0.567 s



5. Registro experimental

Ensayo	Tiempo medido (s)
1	2.050
2	1.940
3	2.100
4	1.980
5	2.030
Promedio	**2.020**

Error (%) = $|t_{teórico} - t_{experimental}| / t_{experimental} \times 100 = |2.001 - 2.020| / 2.020 \times 100 =$ **1.0 %**

Desviación estándar: 0.062 s. Nota según la escala: **100**. Aceleración medida: 0.270 m/s². μ_k experimental: 0.546.

